

Fields

$(F, +, \cdot)$ is $(\mathbb{Z}, +, \cdot)$ -like, distributive

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$(F, \cdot)$ is $(\mathbb{Z}, \cdot)$ -like, distributive, $1 \neq 0$

$$① a + 0 = 0$$

$$② a \cdot 0 = 0 \Rightarrow a \cdot b = 0$$

③ Characteristic = prime

$$\underbrace{1+1+\dots+1}_k = 0, \text{ for } k \text{ prime}$$

④ If $|F| = p$ then F is isomorphic to $\mathbb{Z}_p$

$F[x] = f_0 + f_1x + f_2x^2 + \dots + f_mx^m$
s.t. $f_i \in F, i \leq m$ & $f_m \neq 0$

⑤ $f(x) = 0$, degree = ∞

⑥ Polynomial of degree 1

$$\gcd(a(x), b(x)) = ua + bv$$

if $p(x) | a(x)b(x)$ and $\gcd(p(x), a(x)) = \text{const}$
 $p(x) | b(x)$

⑦ $f(x) \in F[x]$ is irreducible if $f(x)$ is non-constant
 $f(x) \neq p(x)q(x)$

⑧ any $f(x)$ can be written as $p_1(x)p_2(x)p_3(x)\dots$
 p_i is irreducible

⑨ In any field F the multiplicative group $(F \setminus \{0\}, \cdot)$ is at most 1 cyclic subgroup of any given order 'n'
 $c = \{1, \alpha, \alpha^2, \dots, \alpha^{n-1}\}$

Let p be prime, $n \geq 1$
 $\exists$ poly $f(x) \in F[x]$ of degree 'n'

$$\text{Catalan number} = \frac{(2n)!}{(n+1)!} = \sum_{i=0}^{n-1} \binom{n-1}{i} \binom{n-1}{i}$$

Bell numbers: n^{th} partition

$$B_n = \sum_{k=0}^n n! \cdot c_k \cdot B_k$$

$$B(x) \cdot e^x = B'(x)$$

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$\mathbb{Z}_n^* = \{k \mid 1 \leq k \leq n, \gcd(k, n) = 1\}$
 $\phi(n) = |\mathbb{Z}_n^*|$

CRAT

p is prime

$$\gcd(a, p) = 1$$

$$a^{\phi(p)} = 1 \pmod{p}$$

$$d(p)$$

message = m

$$0 \leq m \leq N$$

$$c = \text{enc}(m) = m^e \pmod{N}$$

encrypted message

$$\text{dec}(c) = c^d \pmod{N}$$

$$d = e^{-1} \pmod{\phi(N)}$$

$$\pi(n) = \# \text{ prime } \leq n$$

$$\frac{\pi(n)}{n} \sim \frac{1}{\log n}$$

$$\frac{c}{\log n} \leq \frac{\pi(n)}{n} \leq \frac{C}{\log n}$$

$$\frac{4^n}{2n+1} \leq \pi(n) \leq \frac{4^n}{n}$$

$$\prod_{p \leq n, \text{ prime}} p \leq 4^n$$

$$p \in \left(\frac{2n}{3}, n\right) \Rightarrow p \mid \left(\frac{2n}{3}\right)!$$

$$p \mid (2n)!$$

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- group theory
- Associativity
 - Closure
 - $e \in G$

Abelian $ab=ba$

Isomorphism:

$(G, *)$ similar to $(G', \circ)$

If $\exists$ bijection $f: G \rightarrow G'$

$$f(axb) = f(a) \circ f(b)$$

sub-group: $ab^{-1} \in S, a, b \in S$

$S \subseteq G$

cosets: $\{xa \mid x \in S\} = Sa$

$a \in S, S \subseteq G$

If $a \in S, Sa = S$

Sa = right coset

Total no of distinct cosets = $|G|/|S|$

$|G| \neq 0, |G| = p, p$ is prime

Then G is cyclic

An infinite cyclic group is isomorphic to $(\mathbb{Z}, +)$

SYMMETRY

Symmetry is a distance preserving transformation with maps $TP \rightarrow P$

$$VS = SV^{-1}$$

Let G is group of order $2n$ st.

$$r^n = e$$

$$s^n = e$$

$$VS = SV$$

Then G is isomorphic to D_n

A group G is said to act on set X if $\forall g \in G$ can be associated with a permutation ϕ_g

$$\phi_g: X \rightarrow X, \text{ st.}$$

$$\phi_e(x) = x \quad \forall x \in X$$

$$\phi_{g_1 g_2}(x) = \phi_{g_1}(\phi_{g_2}(x))$$

$$= \phi_{g_1}(\phi_{g_2}(x))$$

$$\text{orb}_x = \{\phi_g(x) \mid g \in G\}$$

$$\text{stab}_x = \{g \in G \mid \phi_g(x) = x\}$$

$$C_x \text{ sub-group}$$

$$|\text{orb}_x| = |G|/|\text{stab}_x|$$

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$$\text{If } a, b \text{ belong to same left coset i.e. } a \sim b$$

$$\text{then } \phi_a(x) = \phi_b(x)$$

$$\text{Total} = \sum \frac{\text{Fix}(g)}{|G|}$$

$$\# \text{ orbits} = \sum \frac{1}{|\text{orb}_x|}$$

$$= \frac{1}{|G|} \sum \text{Fix}(g)$$

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Walk: any v is

tail: $\rightarrow$ directed edges

path: $\rightarrow$ edges vertices

$\text{degree}(v) \geq d \quad \forall v \in V$

then $\exists$ a cycle of length $d+1$

$|N(S)| \geq |S|$, x perfect

M is maximum matching

$\nexists$ augmenting path w.r.t M

Let M^* be maximum matching $|M^*| \geq |M|$

consider $D = (M^* \setminus M) \cup (M \setminus M^*)$

$\text{degree}(v) \leq 2 \quad \forall v \in V(D)$

$\forall k \geq 2, \forall$ graph

If $|G| > (1 - \frac{1}{k-1}) \frac{n^2}{2}$

then G must have a k -clique

If G doesn't have a k -clique then

$$|E(G)| \leq (1 - \frac{1}{k-1}) \frac{|V(G)|^2}{2}$$

Bezout's theorem,

$$\text{gcd}(a, b) = \alpha a + \beta b$$

$$b = a q_1 + r_1$$

$$a = r_1 q_2 + r_2$$

$$r_1 = r_2 q_3 + r_3$$

$$r_2 = r_3 q_4$$

$$r_3 = r_1 - r_2 q_3$$

$$= b - a q_1 - a r_2 q_3$$

$$= b \beta + a \alpha$$